

Distributed Event Streaming Platform Components

Sure! Let's talk about **Event Streaming Platforms (ESP)** in simple terms.

An Event Streaming Platform is like a busy post office that helps send and receive messages (or events) between different places. Imagine you have a group of friends (event sources) who are constantly sending you updates about their day, like where they are or how they're feeling. Instead of each friend sending their updates directly to each other, they send everything to the post office (the ESP). The post office then organizes these messages and makes sure they get to the right friends (event destinations) who want to receive them. This way, everyone stays updated without getting confused!

In this system, events can be different types of information, like the temperature in a room or the speed of a car. The ESP makes it easier to manage all these messages, especially when there are many friends (sources) and many places to send the messages (destinations). It helps keep everything organized and flowing smoothly, just like a well-run post office!

What you will learn



Describe what an Event is



Use the common Event formats



Describe an event streaming platform (ESP)



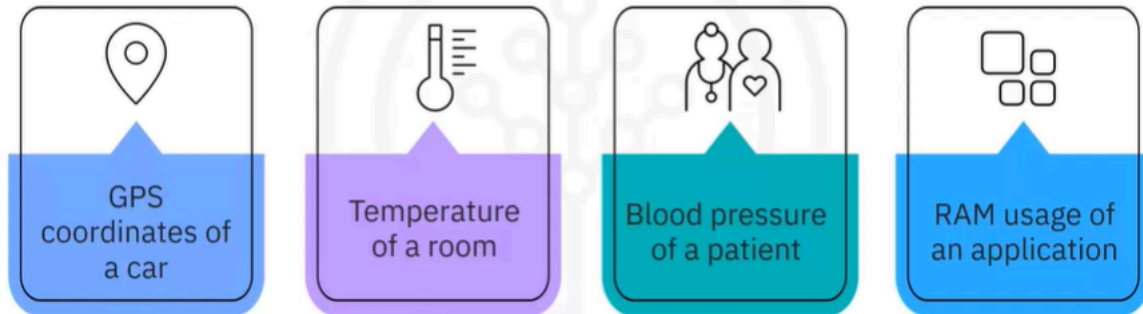
List the main components of an ESP



List the popular ESPs

What is an Event?

An Event describes the entity's observable state updates over time



- An event normally means something worth noticing is happening. In the context of event streaming, an event is a type of data which describes the entity's observable state updates over time. For example, the GPS coordinates of a moving car, the temperature of a room, blood pressure measurements of a patient, or a RAM usage of a running application.

Common Event formats

Three most common formats:

Primitive

"Hello"

A key-value pair

Key: "car_id_1"
Value: (43.82, -79.48)

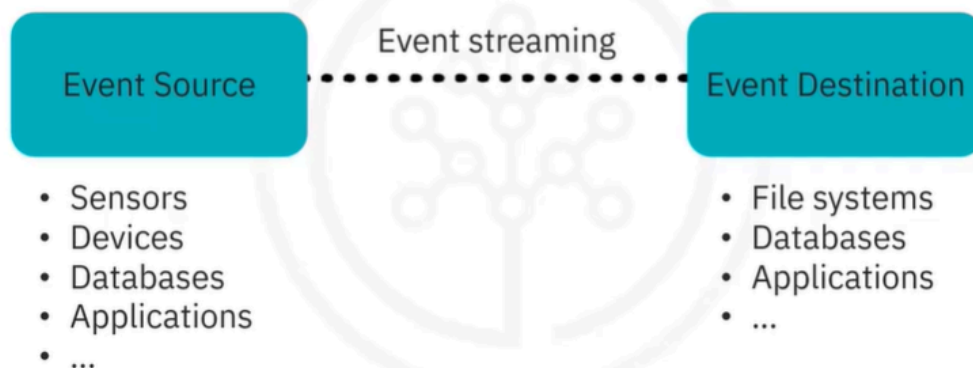
A key-value with a timestamp

Key: "pt001"
Value: (125, 85)
Timestamp: 2024-05-04 12:00

- An event as a special type for data has different formats. Let's have a look at the three most common formats. It can be as a primitive type such as a plain text, number, or date, or an event may be in key-value format, and its value can be a primitive data type, or complex data type like list, tuple, JSON, XML, or even bytes. For example, the GPS coordinates of a car with car_id_1 as a tuple. Also, very often, an event can be associated with a timestamp to make it time-sensitive. For example, the blood pressure of a patient with ID pt001 as a tuple.

One source to one destination

Event streaming from one event source to one destination



- Next, we will have a closer look at event streaming. Suppose we have one event source such as a group of sensors, a monitoring device, a database, or a running application. This event source may continuously generate a large event volume at a short time interval or nearly real time. Those real-time events need to be properly transported to an event destination, such as a file system, another external database, or an application. The continuous event transportation between an event source and an event destination is called event streaming. After all you have learned about ETL so far, you may think that to implement such an ETL process between one event source to one event destination should be straightforward. However, what if we have multiple different event sources and destinations?

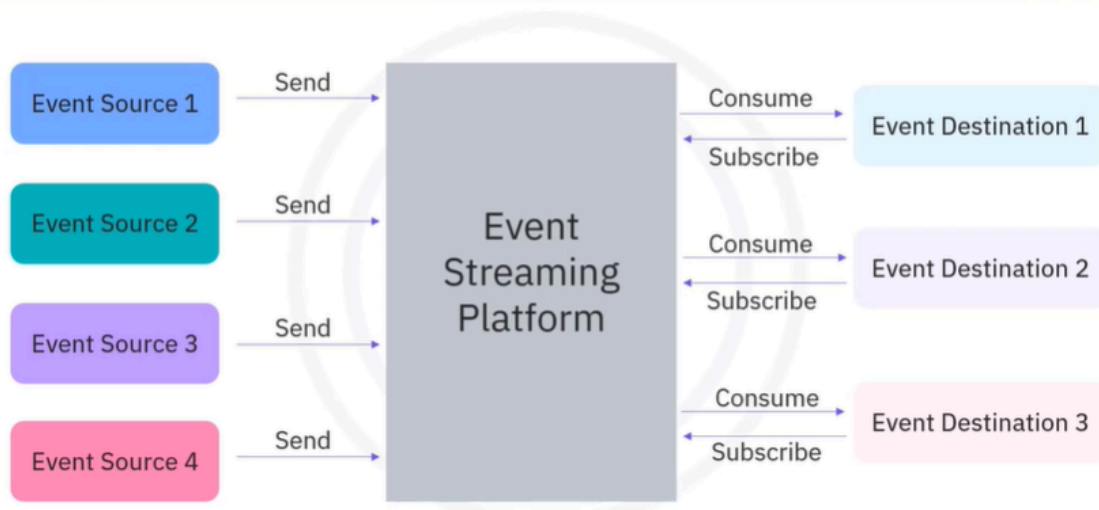
Many sources to many destinations

Event streaming from multiple event sources to multiple destinations



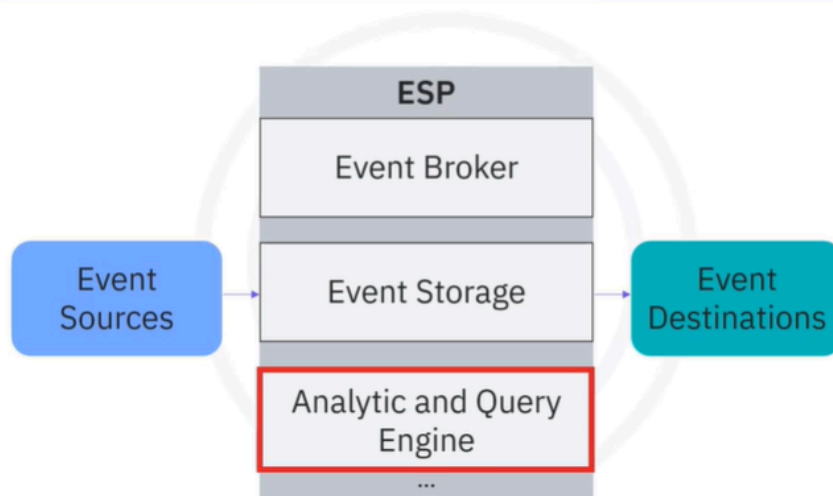
- In a real-world scenario, event streaming can be really complicated with multiple distributed event sources and destinations, as data transfer pipelines may be based on different communication protocols, such as FTP, file transfer protocol, HTTP, Hypertext Transfer Protocol, JDBC, Java database connectivity, SCP, secure copy, and so on. An event destination can also be an event source simultaneously. For example, one application could receive an event stream and process it, then transport the processed results as an event stream to another destination. To overcome such a challenge of handling different event sources and destinations, we will need to employ the event streaming platform.

Event streaming platform (ESP)



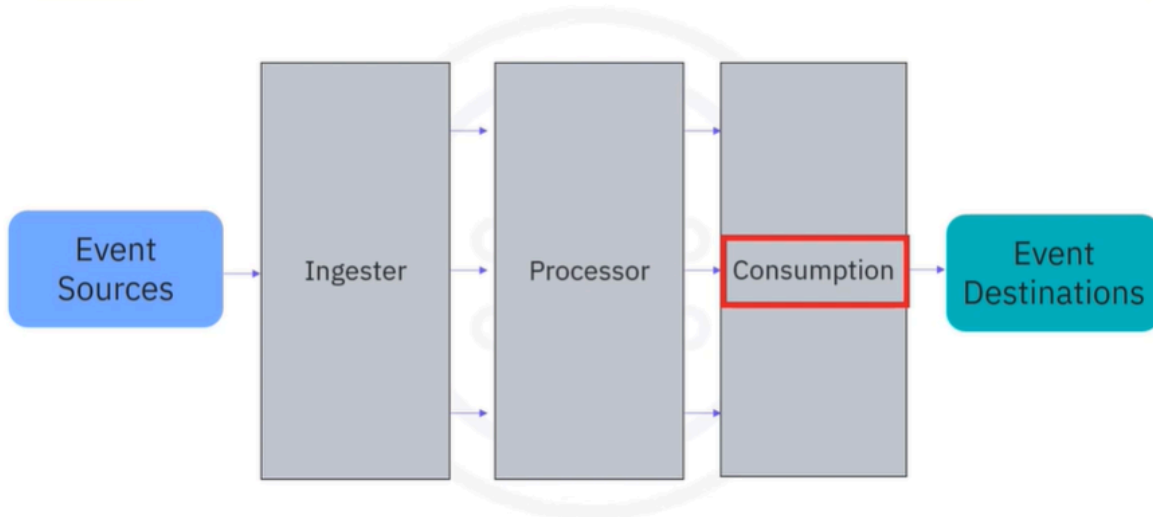
- An ESP acts as a middle layer among various event sources and destinations and provides a unified interface for handling event-based ETL. As such, all event sources only need to send events to an ESP instead of sending them to the individual event destination. On the other side, event destinations only need to subscribe to an ESP and just consume the event sent from the ESP instead of the individual event source.

Common components of an ESP



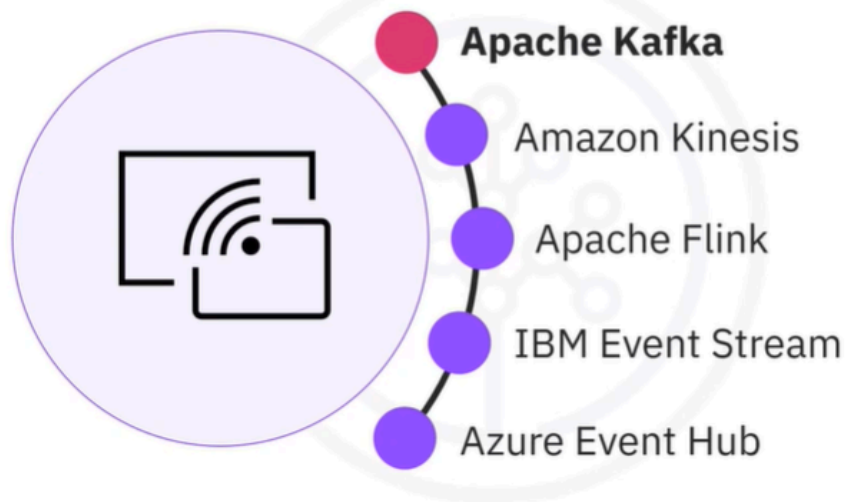
- Different ESPs may have different architectures and components.
- Here we show you some common components included in most ESP systems. The first and foremost component is the event broker, which is designed to receive and consume events. Since it is the core component of an ESP, we will explain it in more detail in the next slide. The second common component of an ESP is event storage, which is used for storing events being received from event sources. Accordingly, event destinations do not need to synchronize with event sources, and stored events can be retrieved at will. The third common component is the analytic and query engine, which is used for querying and analyzing the stored events.

Event broker



- Let's have a look at the event broker, which is the core component of an ESP. It normally contains three subcomponents, ingester, processor, and consumption. The ingester is designed to efficiently receive events from various event sources. The processor performs operations on data such as serializing and deserializing, compressing and decompressing, encryption and decryption, and so on. The consumption component retrieves events from event storage and efficiently distributes them to subscribed event destinations.

Popular ESPs



- There are many ESP solutions, including Apache Kafka, Amazon Kinesis, Apache Flink, IBM Event Stream, Azure Event Hub, and so on. Each has its unique features and application scenarios. Among these ESPs, Apache Kafka is probably the most popular one.

Recap

In this video, you learned that:

- An Event describes the entity's observable state updates over time
- Common event formats include primitive data types, key-value, and key-value with a timestamp
- An ESP is needed, especially when there are multiple event sources and destinations
- The main components of an ESP are Event broker, Event storage, Analytic and Query Engine
- Apache Kafka is the most popular open source ESP
- Other popular ESPs include Amazon Kinesis, Apache Flink, IBM Event Stream, Azure Event Hub